stating the variation (noise) in data. Defined as: 100% × standard deviation / mean.

Commutative property of convolution: Written as: $a[n] \ast b[n] = b[n] \ast a[n]$.

Companding: An "s" shaped nonlinearity allows voice signals to be digitized using only 8 bits instead of 12 bits. Europe uses "*A*" *law*, while the United States uses the *mu law* version.

Complex conjugation: Changing the sign of the imaginary part of a complex number. Often denoted by a star placed next to the variable. Example: if $A = 3 + 2j$, then $A^* = 3 - 2j$.

Complex DFT: The discrete Fourier transform using complex numbers. A more complicated and powerful technique than the *real* DFT.

Complex exponential: A complex number of the form: e^{a+bj}. They are useful in engineering and science because Euler's relation allows them to represent sinusoids.

Complex Fourier transform: Any of the four members of the Fourier transform family written using complex numbers. See *real Fourier transform* for comparison.

Complex numbers: The *real numbers* (used in everyday math) plus the *imaginary numbers* (numbers containing the term j, where $j = \sqrt{-1}$). Example: $3 + 2j$.

Complex plane: A graphical interpretation of complex numbers, with the real part on the x-axis and the imaginary part on the y-axis. This is analogous to the *number line* used with ordinary numbers.

Composite video: An analog television signal that contains synchronization pulses to separate the fields or frames.

Computed tomography (CT): A method used to reconstruct an image of the interior of an object from its x-ray projections. Widely used in medicine; one of the earliest applications of DSP. Old name: CAT scanner.

Continuous signal: A signal formed from continuous (as opposed to discrete) variables. Example: a *voltage* that varies with *time*. Often used interchangeably with *analog signal*.

Contrast: The difference between the bright-ness of an object and the brightness of the background. See *brightness* for comparison.

Converge: Term used in iterative methods to indicate that progress is being made toward a solution ("The algorithm is converging") or that a solution has been reached ("The algorithm has converged").

Convolution integral: Mathematical equation that defines convolution in continuous systems; analogous to the *convolution sum* for discrete systems.

Convolution kernel: The impulse response of a filter implemented by convolution. Also known as the *filter kernel* and the *kernel*.

Convolution sum: Mathematical equation defining convolution for discrete systems.

Cooley and Tukey: J.W. Cooley and J.W. Tukey, given credit for bringing the FFT to the world in a paper they published in 1965.

Correlation: Mathematical operation carried out the same as convolution, except a left-for-right flip of one signal. This is an optimal way to detect a known waveform in a signal.

Cross-correlation: The signal formed when one signal is correlated with another signal. Peaks in this signal indicate a similarity between the original signals. See also *autocorrelation*.

Cutoff frequency: In analog and digital filters, the frequency separating the passband from the transition band. Often measured where the amplitude is reduced to 0.707 (-3dB).

CVSD: Continuously Variable Slope Delta modulation, a technique used to convert a voice signal into a continuous binary stream.

DC: Direct Current. Electrical term for the portion of the signal that does not change with time; the average value or mean. See *AC* for comparison.

Decibel SPL: Sound Pressure Level. Log scale used to express the intensity of a sound wave: 0 dB SPL is barely detectable; 60 dB SPL is normal speech, and 140 dB SPL causes ear damage.

Decimation: Reducing the sampling rate of a digitized signal. Generally involves low-pass filtering followed by discarding samples. See *interpolation* for comparison.

Decomposition: The process of breaking a signal into two or more additive components. Often refers specifically to the *forward Fourier transform*,